Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach tree learning through a repeatable candidate comparison.

Provide complete scripts, distinguish specified policies from fitted predictions and assess the new training set independently.

Keep one habit: Ask which evidence chose the question, which chose the leaves and which only scored the frozen tree.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Fit a **root** question and **leaf prediction**s from **training** examples

Teach two 45-minute sessions. Students fit a **depth**-one **decision tree** by learning **branch** majorities and selecting the question with the fewest total **training** mistakes.

This transparent classroom algorithm is not a claim to reproduce CART or a library default. Explain tuple positions, list comprehension and the leaves dictionary; offer the full paper route for students who need it.

By the end, I can fit a one-question tree using **training** labels and an explicit candidate comparison.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/14, ai-2/15, ai-4/11. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Printed lesson and workbook
- Six **training** row cards and four validation cards
- Two **branch** areas labelled 0 and 1
- Optional Python 3 editor with no packages required

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a category definition using explicit **evidence** and boundary rules

Reference example, not the only acceptable wording: In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room.

Equality at 30 is accepted.

The cutoff was selected using **training** labels. Matching those examples does not establish performance on new examples; validation can expose mistakes and support a baseline comparison.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Types of Classification Error

Training and Testing

Operational Labels

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

Concept to explain

Each question **partition**s possibilities; question order affects efficiency.

Term	Meaning
decision tree	A sequence of branching questions that routes an input to an output at a leaf.
node	A point in a graph or tree representing a state, question or result.
leaf	A final node of a decision tree that supplies an output rather than another question.
partition	One separated portion of a dataset, such as training, validation or test data.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach fitting and prediction as separate stages

> Provide complete candidate and validation scripts before asking learners to interpret their evidence.

State feature meanings and fitting scope

A **decision tree** asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump. In Week10 the Tower1 rule was supplied. Here the class names are still declared in advance, but training labels determine which question is selected and which class **prediction** is stored at each **leaf**.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group’s majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

Supply all six training rows

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear
5	0	0	clear

Row	Near	Tall	Label
6	0	1	blocked

Model the complete candidate comparison

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat split selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Freeze the near tree and collect validation predictions

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

> **Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.**

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

Run the complete same-case baseline comparison

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])

validation = [(1, 1, "blocked"), (0, 0, "clear"),
              (0, 1, "blocked"), (1, 0, "clear")]
feature, leaves = tree
predictions = [leaves[row[feature]] for row in validation]
reference = [row[2] for row in validation]
print("tree", predictions, sum(p == y for p, y in zip(predictions, reference)),
len(reference))
baseline = majority([row[2] for row in rows])
print("baseline", baseline, sum(baseline == y for y in reference), len(reference))
```

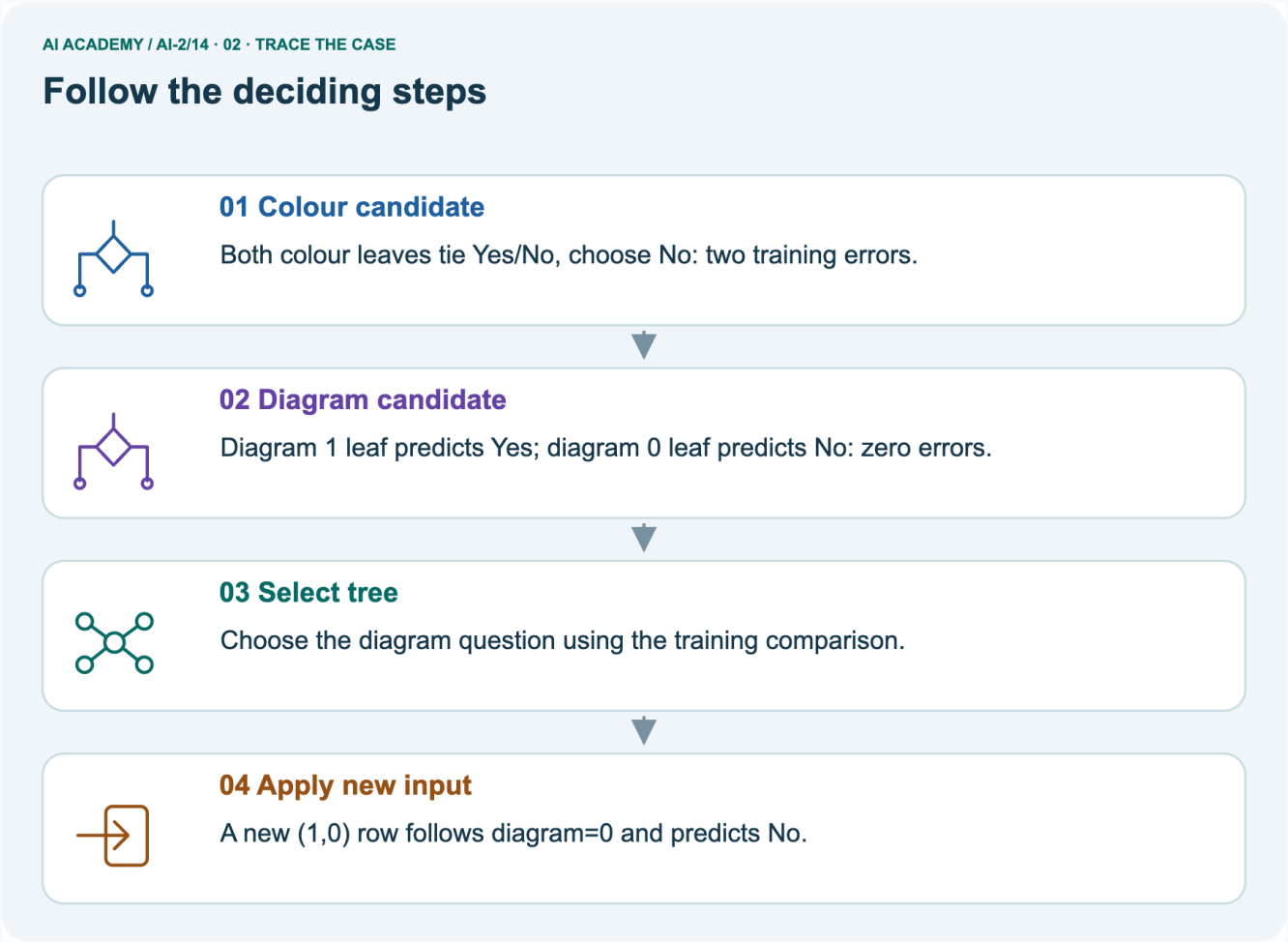
Introduce the separate illustration-label training set

A fictional toy classifier predicts whether a book needs an illustration label.

Training rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

Trace the book candidates and prediction path



Follow the labelled picture with the matching explanation. Open the vector diagram

A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Change a query without refitting

Change: New **input** becomes (0,1).

Prediction: The fitted tree predicts Yes.

Reason: It follows diagram=1; cover colour is not the selected question.

Elicit the picture-based explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Start with 8 animals.

2. Question A leaves groups of 4 and 4; question B leaves 7 and 1.

3. A balanced first split usually reduces later questions.

4. Trace every animal to verify one and only one **leaf**.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

Exact explanation to model

The animal book reaches Short Animal; science reaches Other after the first question.

Explain the added worked example

Key terms

Term	Meaning
Root	The first question asked by a tree
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction
Depth	The largest number of question steps from root to leaf

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

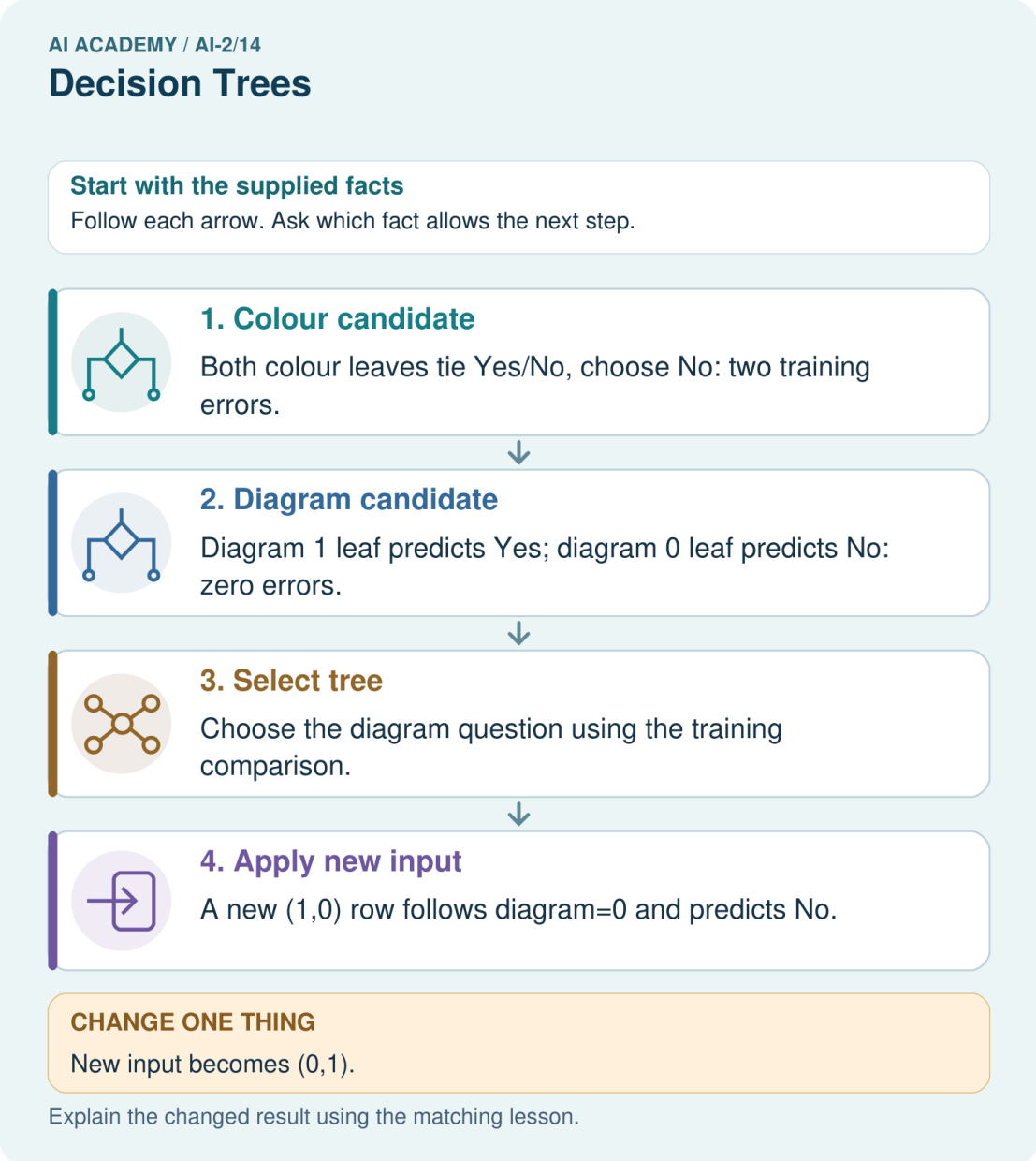
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Decision Trees

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Colour candidate	Both colour leaves tie Yes/No, choose No: two training errors.
2. Diagram candidate	Diagram 1 leaf predicts Yes; diagram 0 leaf predicts No: zero errors.
3. Select tree	Choose the diagram question using the training comparison.
4. Apply new input	A new (1,0) row follows diagram=0 and predicts No.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Expected reasoning and fading help

Teacher check: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear.

Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Fitting** a **decision tree**. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Why does changing colour alone not change this tree's **prediction**?

The fitted tree **branch**es only on diagram_inside; colour was considered but not selected.

Changed-case reasoning: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Write the question to explain: Trace 90-page animal and science books.

Reveal after **prediction**s: The animal book reaches Short Animal; science reaches Other after the first question.

Use the earlier changed question as optional later practice. Reserve Fit a new tree independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: tree **prediction**s blocked,clear,clear,blocked miss rows 3 and 4. Baseline blocked misses rows 2 and 4. Both get 2/4 or 50 percent. Tall cannot affect **prediction** because the stored **root** uses near only.

Exit: the near=1 **leaf** contains three blocked **training** examples, but the whole tree gets 2/4 on this tiny validation set and misses one near=1 validation case. This does not establish reliability on future trips.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve training search versus validation. Introduce root, branch, leaf and depth with one question.
7 to 18 minutes	Sort the six cards by near. Count labels within each branch and assign its majority without using validation.
18 to 32 minutes	Repeat for tall and complete W1 and W2. Add branch mistakes to score the full training set.
32 to 45 minutes	Trace code and the diagram. Predict all four validation rows, complete the guided question and compare the baseline.
Session 2 start to 8 minutes	Retrieve why a leaf label is learned rather than dictated by the feature name.
8 to 20 minutes	Complete W3 and discuss depth limits, tied questions and the assumptions required by the short code.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently before hints. Require branch counts, both candidate scores, learned leaves and validation comparison.
35 to 45 minutes	Use W5 and the exit claim. Give the fresh retry after feedback and record support.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Worked near0: clear2/blocked1→clear,1error; near1: clear0/blocked3→blocked,0errors. Tall0 and tall1 each clear1/blocked2→blocked,1error each. Select near1error versus tall2. Validation predictions blocked,clear,clear,blocked: matches rows1,2 only; training-majority blocked matches rows1,3 only; both2/4. Training score5/6 is distinct. The separate book toy selects diagram with0errors versus colour2, using No on tied leaves. Empty branch handling is not implemented; do not present an empty-list clear result as a designed fallback. Reserve the independent reversed-label training set for W4.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the supplied six-row training set and four-row validation set. Compare the two candidate questions with their majority leaves, save the fitted depth-one tree and trace the separate validation cases. Keep the new independent training set reserved; no eight-card dataset is required.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A near **branch** must predict blocked: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Every feature is used during **prediction**: Only questions in the stored tree are followed. Candidate questions rejected during **fitting** do not affect this tree's **output**.

A deeper tree must work better: More questions can fit **training** detail. Future performance needs separate evaluation; **depth** choices must be documented.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Fit a new tree independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies.

Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

Near=0 predicts blocked from 2 blocked and 1 clear; near=1 predicts clear from 3 clear. Near has 1 error. Both tall groups have 2 clear and 1 blocked, so both leaves predict clear and total errors are 2.

Choose near with reversed **leaf** labels compared with the worked example. Validation **prediction**s clear,blocked,blocked,clear match rows 1,2: 2/4. Baseline clear matches rows 1,3: 2/4. Both are 50 percent but errors differ.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Name the parts and decisions

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

The **root** asks near, with **branch**es 0 and 1. Leaves predict clear for 0 and blocked for 1. **Depth** is one. **Training** labels determined both **leaf** majorities and the error comparison that selected near rather than tall.

W2 Verify the candidate scores

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear.

Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

W3 Separate fitting from prediction

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

The **leaf** was changed using validation **evidence**, so it is no longer the original **training**-fitted tree. Score the original stored leaves on all the same validation rows and record any later development change separately.

Preserve a separate final test for final evaluation.

W5 Audit a tie and a claim

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

It retains near because near is tried first and strict < replaces only for fewer errors. This is a declared tie convention, not **evidence** of better future performance. Separate evaluation and relevant examples are still needed.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

The animal book reaches Short Animal; science reaches Other after the first question.

Additional transfer practice question

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

Reasoned practice answer

First answer yes, second answer no. Follow the **leaf** on that complete path. The first **branch** alone does not determine the final decision.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The first matching feature ends every tree.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Science/100 reaches short science; history/70 reaches other genre. Missing pages cannot answer the length question, so request the value or route to review.

A tree diagram must label **branch**es and leaves, not just list questions.

Explanations for the additional workbook examples

W1 Name the parts and decisions

W2 Verify the candidate scores

W3 Separate fitting from prediction

W4 Fit a new tree independently

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates’ leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

Near=0 predicts blocked from 2 blocked and 1 clear; near=1 predicts clear from 3 clear. Near has 1 error. Both tall groups have 2 clear and 1 blocked, so both leaves predict clear and total errors are 2.

Choose near with reversed **leaf** labels compared with the worked example. Validation **prediction**s clear,blocked,blocked,clear match rows 1,2: 2/4. Baseline clear matches rows 1,3: 2/4. Both are 50 percent but errors differ.

W5 Audit a tie and a claim

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Create a transparent tree for hard constraints before applying preference scores.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use physical sorting before code. Explain row[feature] as selecting a flag and row[2] as reading its label. Supply empty **branch**-count tables.

Extension: explain why an empty **branch** needs a fallback, and why this code would silently return clear if given an empty label list.

Require W4 near error 1, tall error 2, near=0 blocked and near=1 clear, with both validation scores 2/4. Fresh retry: **training** (0,0,clear),(0,1,blocked),(1,0,clear),(1,1,blocked).

Near **branch**es each tie and predict clear, giving 2 errors. Tall=0 predicts clear and tall=1 blocked with 0 errors; choose tall. Validation (0,1,blocked),(1,0,clear),(1,1,clear) gives blocked,clear,blocked: 2/3.

The tied **training** baseline predicts clear and also gets 2/3, with different missed rows. Require the learned feature and both tie policies.

Plan the delayed check

Suggested return points in this level: ai-2/15, ai-2/17, ai-2/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 15: Choose Informative Questions. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week15: Choose Informative Questions. Carry forward candidate definitions, label counts, declared selection criteria and separate validation results. Teacher reference: https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html (reviewed2026-09-27); its library criteria differ from this stated classroom mistake-count procedure.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Fit a **root** question and **leaf prediction**s from **training** examples

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Reasoned practice answer: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors.

Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear. Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Check every learner before discussion: Use the worked case for **Fitting** a **decision tree**. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: New **input** becomes (0,1). Require a **prediction** and a reason before comparing. Expected result: The fitted tree predicts Yes. Reason: It follows diagram=1; cover colour is not the selected question.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The **branch**-boundary and indexing lessons deliberately include failure cases.
</details>

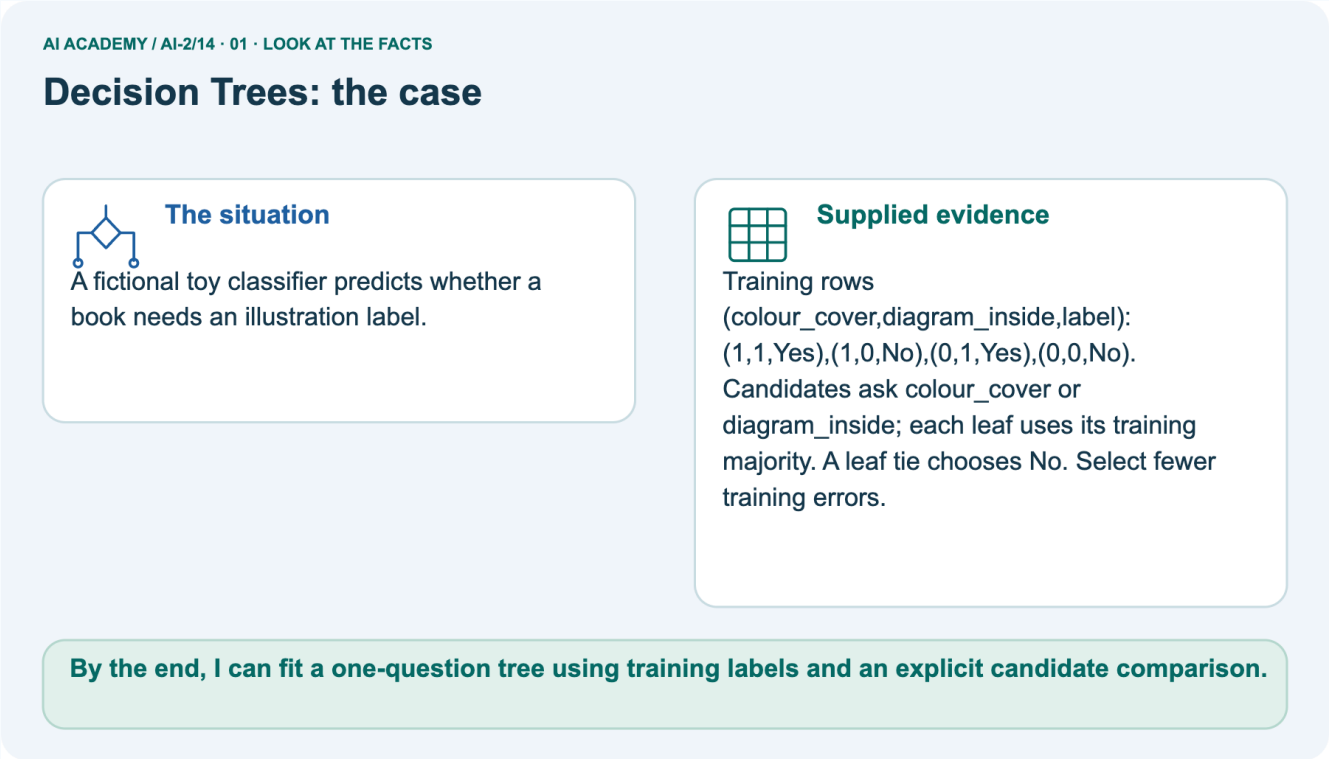
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Decision Trees: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

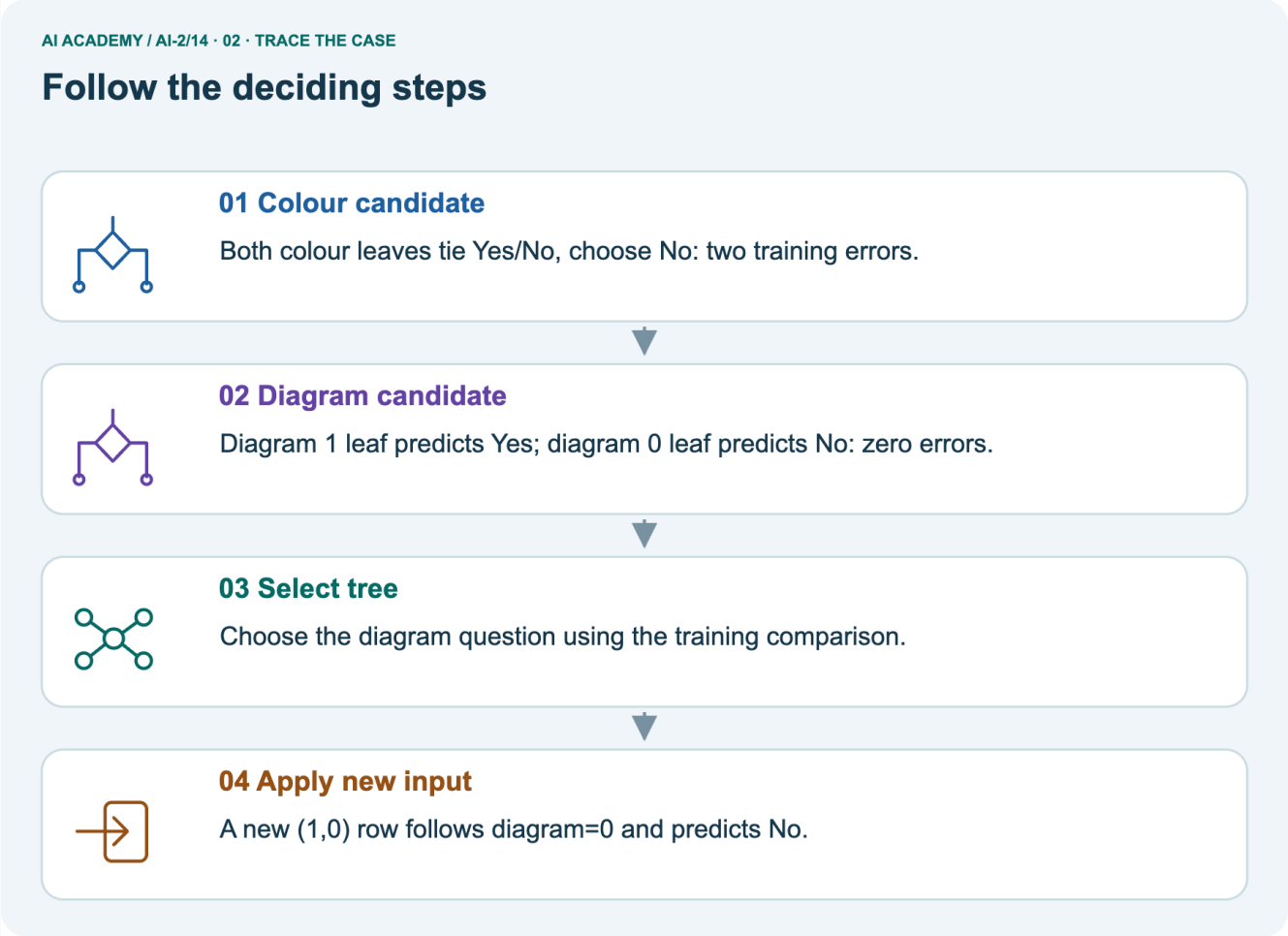
A fictional toy classifier predicts whether a book needs an illustration label. Supplied facts: **Training** rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



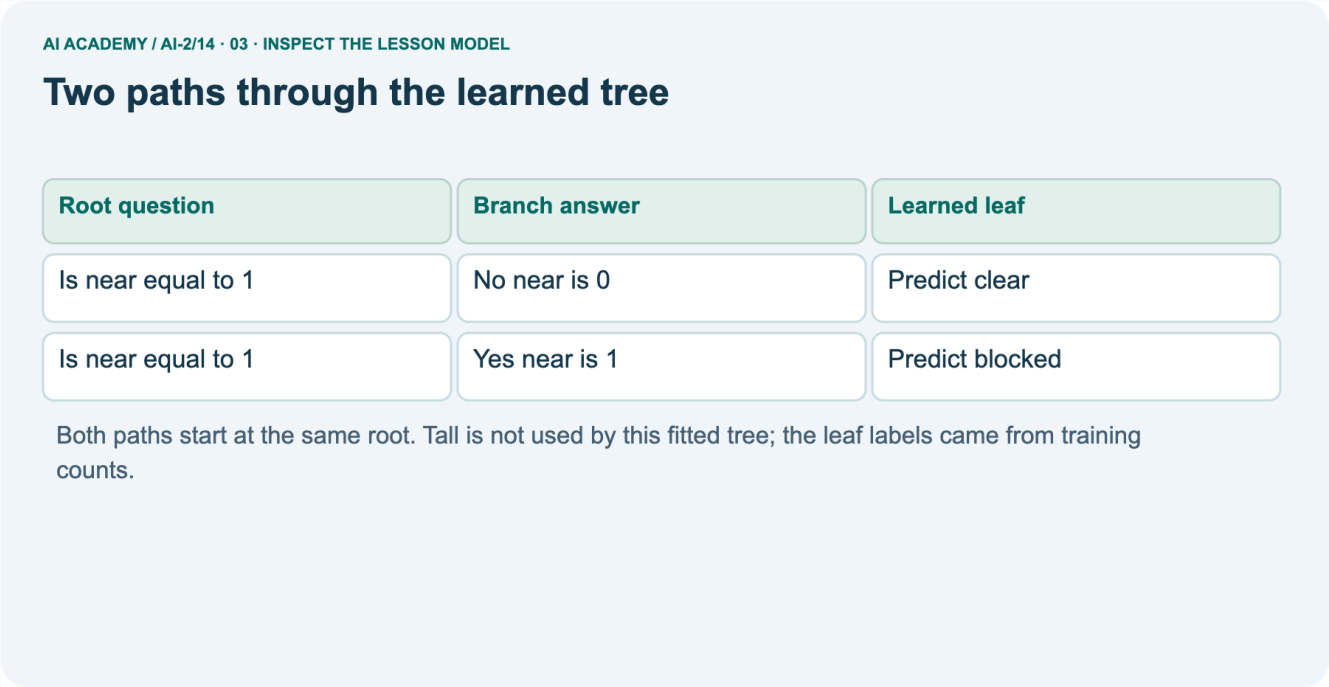
Follow the labelled picture with the matching explanation. Open the vector diagram

Colour candidate: Both colour leaves tie Yes/No, choose No: two **training** errors. → Diagram candidate: Diagram 1 **leaf** predicts Yes; diagram 0 **leaf** predicts No: zero errors. → Select tree: Choose the diagram question using the **training** comparison. → Apply new **input**: A new (1,0) row follows diagram=0 and predicts No.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two paths through the learned tree



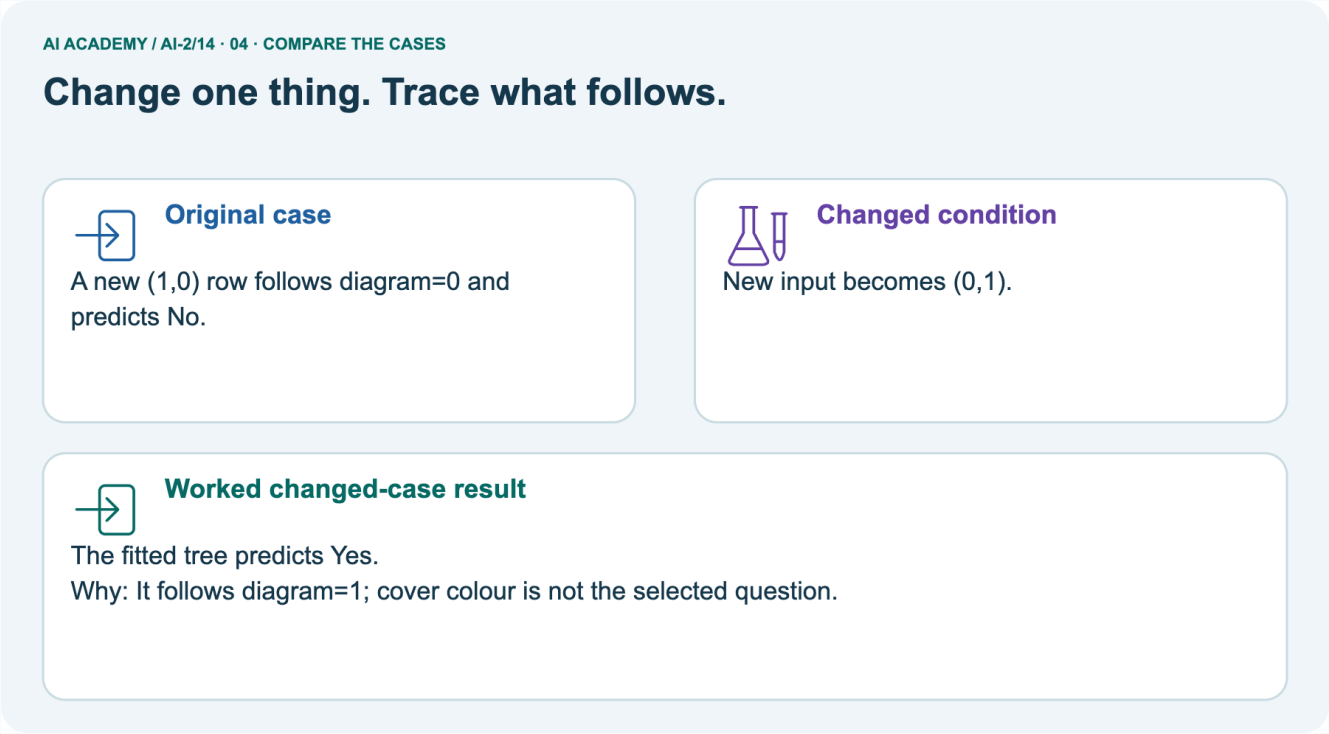
Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A new (1,0) row follows diagram=0 and predicts No. Changed condition: New **input** becomes (0,1). Expected result: The fitted tree predicts Yes.

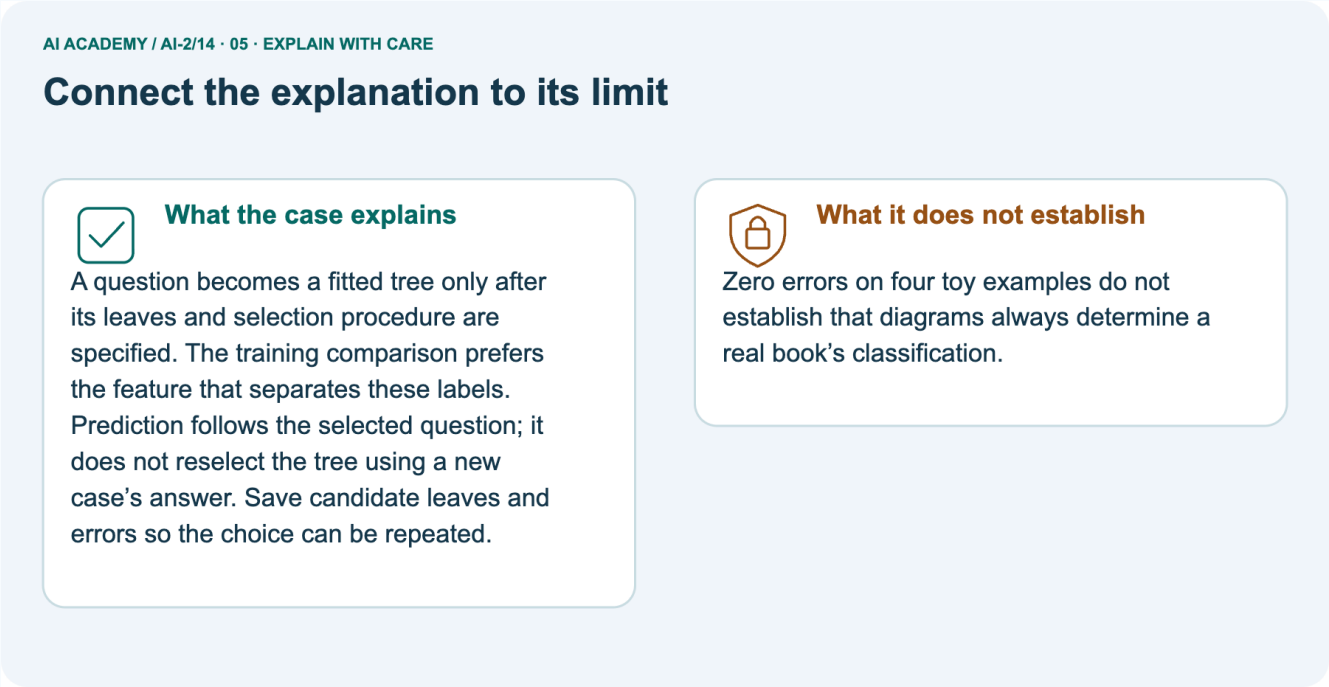
Reason: It follows diagram=1; cover colour is not the selected question.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: It follows diagram=1; cover colour is not the selected question.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

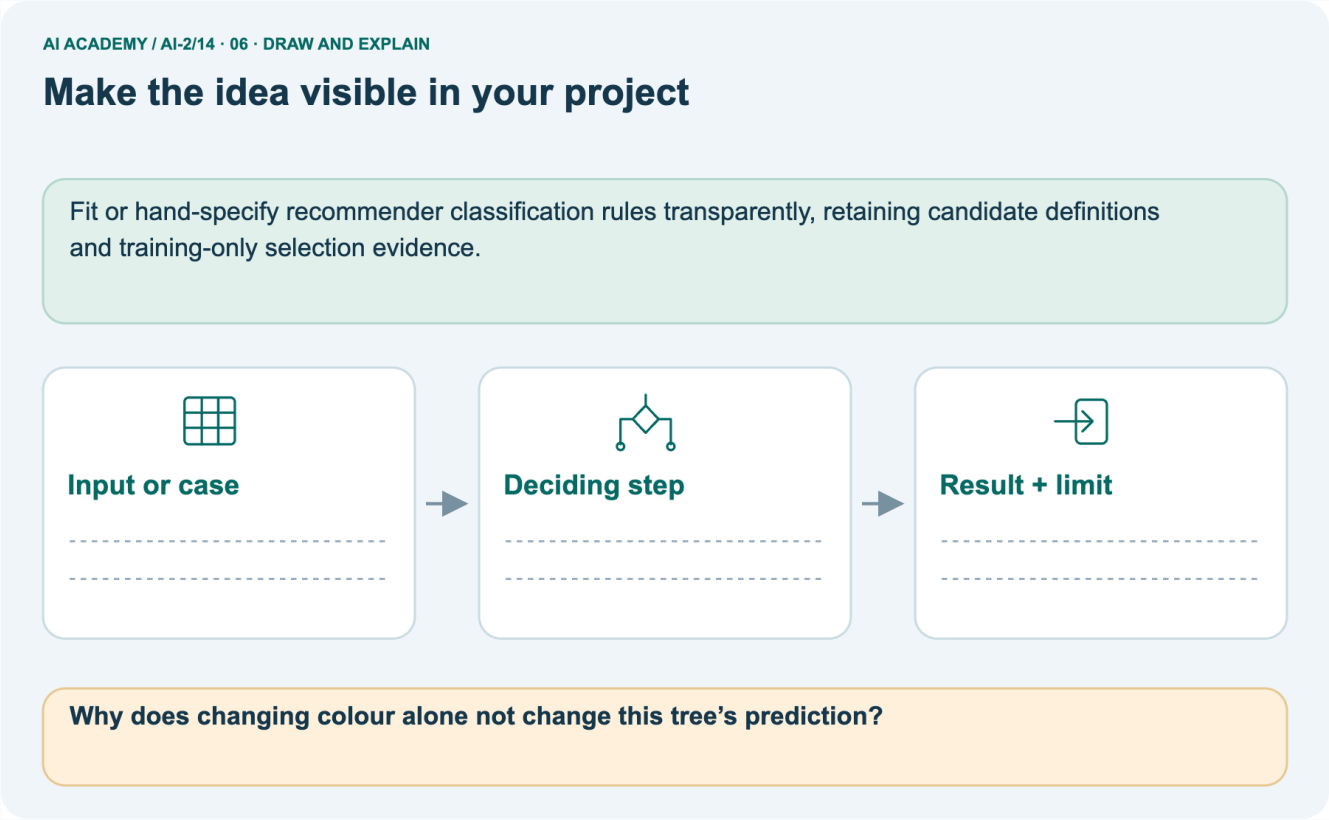
Limit: Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Why does changing colour alone not change this tree's **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The fitted tree **branch**es only on diagram_inside; colour was considered but not selected.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Decision Trees: the case

02. Follow the deciding steps

03. Two paths through the learned tree

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.